

LoRa UAV Ground Control Station

Pinout Reference Document

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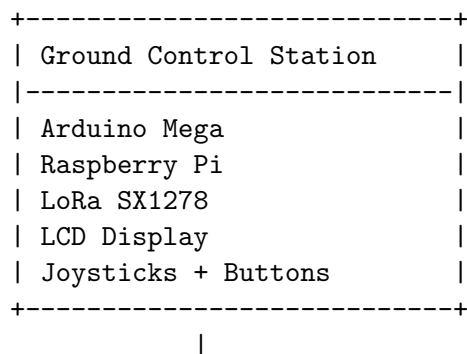
DIY Long Range UAV Ground Control System
Arduino Mega + Raspberry Pi + LoRa + LCD Interface

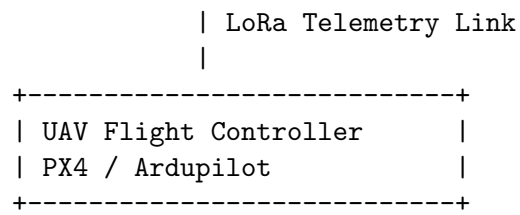
1. Project Overview

This document contains the complete pinout reference for the DIY UAV Ground Control Station (GCS) developed using:

- Arduino Mega 2560
- Raspberry Pi
- LoRa SX1278 Telemetry Module
- LCD HDMI Display
- Joystick Control System
- Push Buttons
- GPS Module

2. System Architecture





3. LoRa SX1278 Connections

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VCC	3.3V
GND	GND
NSS	D53
SCK	D52
MOSI	D51
MISO	D50
RESET	D9
DIO0	D2

4. Joystick Connections

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VRX	A0
VRX	A1
SW	D7
VCC	5V
GND	GND

5. Button Connections

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ARM	D22
MODE	D23
RETURN HOME	D24
TAKEOFF	D25
LAND	D26

6. LCD Display Connections

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HDMI	HDMI Port
Touch USB	USB Port
Power	5V 3A

7. Power Distribution

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Arduino Mega	5V
LoRa SX1278	3.3V
Raspberry Pi	5V 3A
LCD Display	5V
Cooling Fan	5V

8. Telemetry Configuration

- Frequency: 433 MHz / 868 MHz / 915 MHz
- Spreading Factor: 7
- Bandwidth: 125 KHz
- TX Power: 20 dBm
- Communication Protocol: LoRa

9. Recommended Features

- Long Range Telemetry
- Real-time UAV Monitoring
- Mission Planning
- GPS Navigation
- Return-to-Home
- FPV Video Integration
- Battery Monitoring

10. Safety Notes

- Always use a common GND.
- Never power LoRa SX1278 using 5V.
- Use voltage regulators for stable operation.
- Install external antennas for long-range communication.
- Add cooling for Raspberry Pi operation.

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